# Ananya Panayappan

IBDP Student · 2027 · Bengaluru, India

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## Education

### IB Diploma Programme · Candor International School, Bengaluru

Expected June 2027

I earned 43/45 in IBDP1, with 7s in Mathematics AA HL, Economics HL, and Computer Science HL. For my Mathematics Extended Essay, I analysed the mathematical underpinnings, assumptions, and limitations of Geometric Brownian Motion (GBM) and its validity against S&P 500 stock price data. For my Economics Internal Assessment, I evaluated Pigouvian taxation, imported inflation, and trade protection.

### IGCSE Distinction · June 2025

I ranked first in a cohort of 60 with eight A* grades and earned ICE Distinction. I scored 98 PUM in Additional Mathematics and 97 PUM in Economics and Computer Science.

Additional university study: I studied financial mathematics, calculus, statistics, college algebra, and microeconomics. My full course details appear under Additional coursework below.

## Standardized testing

I scored 1570 on the SAT in May 2026: 800 in Mathematics and 770 in Reading and Writing. I scored 1450 on the PSAT NMSQT. I earned 5s in AP Microeconomics and AP Calculus BC, with a Calculus AB subscore of 5, in May 2026.

## Research & internships

### Pioneer Academics · Student Researcher

Grades 10-11

I developed a quantitative portfolio-resilience framework for Indian markets during COVID-19 and demonetisation, focusing on stochastic processes, probability, and quantitative finance. I earned an A and ranked in the top 5% of undergraduate scholars evaluated by NYU Professor Hesam Oveys. My paper was accepted by the International Journal of High School Research, with publication due in November 2026.

### Independent Research · Financial Econometrics

Grade 11-ongoing

I am investigating India VIX and macroeconomic drivers through a regression model of global volatility transmission to Indian markets, mentored by IIMA Professor Vineet Virmani.

### Dyota Solutions · Quantitative Finance Intern

Grades 11-12

I analysed 12 equity fund categories against benchmarks, simulating lump-sum and SIP returns using CAGR, XIRR, and win rates under CEO Kaushik Ramachandran.

### Mathematics Extended Essay · Geometric Brownian Motion

Grades 11-12

I derived GBM using Itô's Lemma and evaluated its assumptions against five years of S&P 500 data, analysing log returns, volatility, and overnight price movements. I identified heavy-tailed returns and volatility clustering and assessed alternatives including GARCH and jump-diffusion.

## Projects

### PocketSmart · Founder & Lead Developer · Grade 11

I conceptualised, developed, and launched a personal-finance app for teenagers on Google Play. I designed allowance tracking and automated savings to encourage more deliberate financial decisions and greater investment. I built the app to help teenagers track allowances, reflect on spending, set savings goals, and evaluate opportunity costs. Get PocketSmart on Google Play

### Mitra Jyothi · Volunteer Project Lead · Grades 9-12

I built a Braille validation tool for the Braille press, achieving reported accuracy above 70%, easing bottlenecks in accessible textbook production, and saving 2,000+ annual proofreading hours. I contributed to an initiative serving 4,000+ students across 77 schools in India.

### OptimalHire · AI Recruitment Application · Grades 11-12

I built a recruitment web app using React.js, Python Flask, MySQL, BERT, and a Siamese network to rank job and candidate matches by semantic similarity.

## Leadership & entrepreneurship

**Mysteraskin · Co-Founder, Finance & Creative Lead · Grades 10-12**

I lead financial planning, budgeting, cash-flow management, and analysis while developing product concepts for a handmade skincare venture. My work at Mysteraskin

**Project Flow · Founder & Digital Media Coordinator · Grades 10-12**

I spearheaded menstrual health education across 22 schools, organised sustainable menstrual product drives, and built the project website. Visit Project Flow

**Math Club, Candor International School · Founder · Grades 11-12**

I founded a math club for Grades 6-9 with applied game theory and graph theory activities and organised the schoolwide MegaMatics competition.

**CanMUN & Aletheia · Executive Organising Team · Grades 10-11**

I helped organise Candor MUN and an interschool sports event, handling registrations and security for 1,000+ attendees.

## Additional coursework

I earned an A (4.0/4.0 GPA) in Financial Mathematics through concurrent enrollment at Johns Hopkins University in Grade 10. At Columbia University in Grade 10, I studied Thinking and Problem Solving: Math in the Real World, covering graph theory, combinatorial modelling, and game theory.

In Grade 11, I completed Wharton's Essentials of Finance, pitched an investment proposal, and worked with a team on fundamental valuation of Costco stock. Through Arizona State University ULC in Grades 9-11, I earned As in College Algebra, Precalculus, Calculus for Engineers I and II, Statistics, and Microeconomics.

My Coursera coursework in Grades 9-12 includes Financial Markets (Yale, Robert Shiller), Programming for Everybody Specialisation (University of Michigan), The Nuts and Bolts of Machine Learning (Google), Enjoyable Econometrics (Erasmus University), and Singing Popular Music (Berklee College of Music).

## Honors & awards

I earned international rank 4 in the SOF International Mathematics Olympiad (2021) and consecutive school gold medals (2019-2024). I placed first in Epsilon (2023 and 2024) and received Wharton's Best Rationale Award (2026).

I placed first in Eton College's startup elevator pitch presentation (2023); Binary Brawl (2023), SNIS TechFest (2023), and Candor TechFest (2024) hackathons; Candor Entrepreneurship Day's Harry Potter soap stall (2023); and Candor Verbattle (2023).

## Skills

My mathematics and statistics skills include mathematical modelling, stochastic processes, game theory, graph theory, statistical and regression analysis, calculus, probability theory, and combinatorial modelling.

My economics skills include data-driven economic research, quantitative economic analysis, econometrics, statistical modelling, microeconomics, and macroeconomics. I work with statistical machine learning, natural language processing (NLP), BERT, Siamese networks, React.js, Python, Flask, and MySQL.

## Interests & initiatives

Through the Pioneer Alumni Network, I shared insights and guidance on the research process. As Chief Editor of Meraki, my school magazine, I wrote creative, formal, and informative pieces. I also enjoy vocal music, keyboard, original music composition, baking, and pastry techniques.

## Technical qualifications & languages

I became a Certified Game Developer with Deep UI/UX Interface through WhiteHat Jr (2020) and a Certified Web Developer Level 2 through GiveMeFive (2022). I speak English, Tamil, and Hindi.